

Overview

Welcome to CSC209H, Software Tools and Systems Programming. The course is intended to introduce software development techniques and tools in a Unix-style environment. It focuses on the interface between user programs and the operating system and is the gateway to advanced systems courses like operating systems (CSC369), networks (CSC358), information security (CSC347), and databases (CSC343).

We will study the memory model, system calls, and systems structures including files, sockets, and processes. We'll use the C language throughout the course, so by the end of the term, you should be comfortable developing programs in C that interact with the operating system. You should also be familiar with common Unix-style tools like `make`.

Contact Information

Instructors	Furkan Alaca	Ilir Dema	Andrew Petersen
Office	DH3074	DH 3013	DH3096
Office Hours	Tue. 11:00-12:00 Thu. 15:00-16:00	TBA	MW 10:00-12:00
Lectures	LEC0101 LEC0102	LEC0103	LEC0104
Tutorials	Weekly, by ROSI registration (DH labs)		

Website and Discussion Board

The course website contains important information: links to assignments, lecture materials, the discussion board, PCRS, and MarkUs. The website is available through Quercus, or directly via <https://mcs.utm.utoronto.ca/~209>

The discussion board should be your first stop for CSC209 information and is *required* reading. We will post important updates and announcements there, so make sure to visit the board often. In addition, each of you can post questions (and answers!) so the discussion board will be the fastest way to get help with course material or to obtain an answer to an administrative question.

Please do not post *solution code* on the board. If you have a question about assignment code, please generate new code to demonstrate your question, rather than posting the assignment code. Alternately, you can phrase your question abstractly, without posting the code itself.

The board will become very active, so to make it easier for everyone to find answers to their questions, please use good forum etiquette. Use informative titles for your posts, so that people can find relevant information. Read the posts already on the board before posting a question so that you don't post a duplicate. Finally, be professional in tone and behaviour. If your post duplicates previous questions or violates forum etiquette, it may be removed to keep the board organized.

Email

Please use the discussion board to ask general course-related questions. If you have a personal issue, such as needing to report an illness or discussing an alternate test date, please email Prof. Petersen at andrew.petersen@utoronto.ca. Email must be sent from a U of T address; if not, you may not receive a response.

You may also use the Anonymous Feedback feature on the website to provide feedback. Since the sender cannot be determined, comments sent through the feedback system are considered public and may receive a response at the beginning of class or on the discussion board.

Materials

- Textbooks: This course has no required text: we primarily use videos and exercises linked from the course webpage to prepare for lecture. However, you may find it useful to have good C and syscall references handy, and we will post optional readings from these texts:
 - King. *C Programming: A Modern Approach*. W. W. Norton and Company, 2008.
 - Kerrisk. *The Linux Programming Interface*. No Starch Press, 2010.
- Software: The software required for this course has been installed in the Linux Labs (DH2020/2026). You can access these systems via ssh. (Each lab machine is named `dhXXXXpcYY.utm.utoronto.ca`, where XXXX is the room number and YY is a number between 01 and about 30.) If you wish to install it on a personal machine, we strongly recommend installing your favorite flavor of linux on a virtual machine, so you can back it up and restore if necessary. The course staff will not be able to help you install software on your own machines.

Accessibility

The University of Toronto and your instructors are committed to accessibility. If you require accommodations, or there is anything course-related we can do to help, please talk to us.

Work	Weight	Comment
Lecture Prep	5%	Due weekly, Sunday at 10 p.m.
Class Participation	3%	Ongoing, in class
Lab Exercises	10%	Due weekly, Friday at 10 p.m.
Assignment 1 (C Basics and Pointers)	5%	Mon, Jan 27 @ 10 p.m.
Assignment 2 (System Calls and Dynamic Memory)	9%	Wed, Feb 12 @ 10 p.m.
Assignment 3 (Processes)	9%	Mon, Mar 9 @ 10 p.m.
Assignment 4 (Communication)	9%	Mon, Mar 30 @ 10 p.m.
Midterm	10%	Mon, Mar 2 @ 7:10 p.m. in IB110
Final Exam	40%	Minimum of 40% required

Marking
Scheme

Lecture Prep
and
Lab Exercises

Assignments

Research consistently shows us that students remember only a small fraction of what we present in lecture. It is not easy to make sense of material that you see for the first time in the first half-hour of a fast-paced lecture environment, let alone to stay focused for 2 hours. It's also important to space out your studying (spaced repetition). To prime you for what we will discuss, you will view a set of videos and complete exercises by Monday morning, before the first lectures of the week. (In some weeks, the videos will be supplemented or replaced by recommended readings and an activity.) These are the "Preparation" exercises hosted on PCRS: <https://pcrs.utm.utoronto.ca/209>.

Many of the lab exercises are also hosted online. Nevertheless, we strongly recommend that you attend tutorial to start the exercise. Your tutorial TA will present a brief introduction (or review) of the material and will be available to answer questions as you begin to work on the exercise.

Assignments are to be completed individually (no groups). All assignments are to be electronically submitted by 22:00 on their due date. The results of our tests will normally be mailed to you within 24 hours. You may choose to resubmit, fixing any errors detected by our tests or to submit a late submission; these "resubmissions" will be accepted up to 48 hours after the initial submission date. We will mark both the initial submission and the resubmission. We will apply a **10% deduction** to the resubmission mark and then assign you the higher of the two marks.

No late assignments are accepted. (We have resubmissions if you need more time.) In the event of an illness or other catastrophe, have your doctor fill out an official U of T medical certificate and contact Prof. Alaca as soon as possible. It is always easier to make alternate arrangements before the due date.

As your assignments are submitted electronically and will often be tested using an automated testing program, you must follow the submission instructions exactly. (Be particularly careful with the format of any output generated.) Any program that does not compile on a CS lab machine or which crashes (seg fault or bus error) will receive a grade of 0. Check your submission carefully; verify that you have submitted exactly the files you intended to submit and that they compile on a lab machine.

Midterm and
Final Exam

There is one midterm, **held on Monday, March 2 (7:10-8:30 p.m.) in IB110**. Please let us know as soon as possible if you have a conflict or are ill, so we can schedule an alternate time that you can attend.

The final exam is comprehensive and takes place after classes are over. To pass the course you must receive at least 40% on the final exam. The highest mark you will receive if your grade on the final exam is less than 40% is 47%. The final exam schedule is not yet available but will be posted in the middle of February. We will make an announcement once the registrar's site has final exam information.

Academic
Offenses

All of the work you submit must be done by you alone, and your work must not be submitted by someone else. Plagiarism is academic fraud and is taken very seriously. The department uses software that compares programs for evidence of similar code. Please read the Rules and Regulations from the U of T Calendar (especially the Code of Behaviour on Academic Matters):

<http://www.governingcouncil.utoronto.ca/policies/behaveac.htm>

Please don't cheat. It is unpleasant for everyone involved, including us. Here are a couple of general guidelines to help you avoid plagiarism:

- Never look at another student's assignment solution. Never show another student your assignment solution. This applies to all drafts of a solution and to incomplete and even incorrect solutions.
- Keep discussions with other students focused on *concepts* and *examples*. Never discuss assignments before the due date with anyone but your instructor and your TAs.
- Do not look for assignment solutions or partial solutions on the web. If you use the web as a reference and find utility code (general code that is not directly related to the task required by the assignment) that you want to use, please check with your instructor first, and then please cite the source by placing a comment above the copied code that describes the code being copied and that provides a link to the original source.